Derivatives: Contracts, Models, and Markets · Technical Note 25

The Risky Annuity and the Protection Leg, Exactly

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Section 25.3 of Derivatives: Contracts, Models, and Markets writes the two legs of a credit default swap as sums over premium dates, with a $1/2$ *standing in for the coupon accrued at default, and shows that under flat curves the par spread collapses to a hyperbolic tangent. Example 25.1 then finds that paying the protection when default actually happens moves the par spread from 249.998 to 250.94 basis points, and a quarterly contract at a 1% hazard rate prices 0.376% above the credit triangle (Example 4.1). This note does the integrals those results compress. With the risk-free rate and the hazard rate constant on each accrual period, both legs have closed forms; with flat curves the sums are geometric and the par spread is independent of maturity; and the three conventions the chapters use—exact payment at default, the half-period rule, and payment at the period end—are placed side by side, so that it is clear which of them gives the tanh formula, which gives the 0.376%, and why the credit triangle is the limit of all three.*

1 SETUP

Coupon dates $0 = T_0 < T_1 < \cdots < T_n = T$, accrual fractions $\Delta_k = T_k - T_{k-1}$, unit notional, a coupon of one per annum, and the accrued coupon paid at default. On each accrual period $(T_{k-1}, T_k]$ the risk-free rate and the hazard rate are constant, $r(t) = r_k$ and $h(t) = h_k$, and it is convenient to write $\mu_k = r_k + h_k$ for their sum. Nothing requires the two to be constant across periods; the piecewise-constant hazard is exactly what Section 24.3's bootstrap produces.

The object everything is built from is the *risky discount factor*, the riskless discount factor times the survival probability,

$$D(t) = P(t)\,Q(t) = \exp\Big(-\int_0^t \big(r(s) + h(s)\big)\,ds\Big), \tag{TN25.1}$$

which is Section 24.8's observation that a defaultable cash flow is discounted at $r + h$. This D already contains the survival probability, unlike the riskless $D(t)$ of Section 25.3: the chapter's $D(t_i)\,Q(t_i)$ is D_i here, its $D(t_i)$ is P_i, and its λ is h. Writing $D_k = D(T_k)$,

$$\begin{aligned} D_0 &= 1, \qquad D_k = D_{k-1}\,e^{-\mu_k \Delta_k}, \\ D(t) &= D_{k-1}\,e^{-\mu_k (t - T_{k-1})} \quad \text{for } t \in (T_{k-1}, T_k]. \end{aligned} \tag{TN25.2}$$

The density of the default time is $-dQ(t) = h(t)\,Q(t)\,dt$, so a payment of $g(t)$ made at the default time has present value $\int g(t)\,h(t)\,D(t)\,dt$.

2 THE TWO LEGS WITHOUT APPROXIMATION

The risky annuity. The premium leg per unit of coupon collects Δ_k at each T_k the issuer survives to, and the accrued fraction $t - T_{k-1}$ if default arrives at t inside the

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period:

$$A = \underbrace{\sum_{k=1}^{n} \Delta_k \, D_k}_{\text{coupons on survival}} + \underbrace{\sum_{k=1}^{n} \int_{T_{k-1}}^{T_k} (t - T_{k-1}) \, h(t) \, D(t) \, dt}_{\text{accrued coupon at default}}. \tag{TN25.3}$$

The accrual integral is elementary once (TN25.2) is substituted. With $u = t - T_{k-1}$,

$$\begin{aligned} \int_{T_{k-1}}^{T_k} (t - T_{k-1}) \, h_k \, D(t) \, dt &= h_k D_{k-1} \int_0^{\Delta_k} u \, e^{-\mu_k u} \, du \\ &= \frac{h_k D_{k-1}}{\mu_k^2} \Big[1 - (1 + \mu_k \Delta_k) \, e^{-\mu_k \Delta_k} \Big], \end{aligned} \tag{TN25.4}$$

so the risky annuity is

$$A = \sum_{k=1}^{n} \left[\Delta_k D_k + \frac{h_k D_{k-1}}{\mu_k^2} \Big(1 - (1 + \mu_k \Delta_k) \, e^{-\mu_k \Delta_k} \Big) \right]. \tag{TN25.5}$$

Because $D_{k-1} e^{-\mu_k \Delta_k} = D_k$, the same sum can be written

$$A = \sum_{k=1}^{n} \left[\frac{r_k}{\mu_k} \, \Delta_k D_k + \frac{h_k}{\mu_k^2} \big(D_{k-1} - D_k \big) \right], \tag{TN25.6}$$

which separates each period into a survival coupon weighted by r_k / μ_k and a default term proportional to the drop in the risky discount factor across the period.

The protection leg. With recovery R the buyer receives $1 - R$ at the default time if it falls before T:

$$V = (1 - R) \int_0^T P(t) \, \big(- dQ(t) \big) = (1 - R) \sum_{k=1}^{n} \int_{T_{k-1}}^{T_k} h_k \, D(t) \, dt. \tag{TN25.7}$$

The same substitution gives

$$\int_{T_{k-1}}^{T_k} h_k \, D(t) \, dt = h_k D_{k-1} \int_0^{\Delta_k} e^{-\mu_k u} \, du = \frac{h_k}{\mu_k} \big(D_{k-1} - D_k \big),$$

so

$$V = (1 - R) \sum_{k=1}^{n} \frac{h_k}{\mu_k} \big(D_{k-1} - D_k \big). \tag{TN25.8}$$

Each term is the loss times the risky-discounted probability of default in period k; the factor h_k / μ_k is the share of the drop $D_{k-1} - D_k$ that is due to default rather than to discounting.

The par spread. Setting the legs equal,

$$s = \frac{V}{A} = \frac{(1 - R) \sum_k \dfrac{h_k}{\mu_k} \big(D_{k-1} - D_k \big)}{\sum_k \left[\dfrac{r_k}{\mu_k} \, \Delta_k D_k + \dfrac{h_k}{\mu_k^2} \big(D_{k-1} - D_k \big) \right]}. \tag{TN25.9}$$

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Both legs share the block $\frac{h_k}{\mu_k}(D_{k-1} - D_k)$; the accrual part of the annuity is $1/\mu_k$ times that block less $\frac{h_k}{\mu_k}\Delta_k D_k$. Two remarks follow. If $\mu_k = 0$ on some period the expressions are to be read as limits: the accrual term tends to $h_k D_{k-1}\Delta_k^2/2$, the protection term to $h_k D_{k-1}\Delta_k$, and $D_k = D_{k-1}$. And the *half-period rule* of Section 25.3 replaces the accrual term by $\frac{1}{2}\Delta_k P_k(Q_{k-1} - Q_k)$, which is what one gets by assuming default lands at the midpoint of the period and the accrued coupon is paid at its end (TN25.16); (TN25.5) is what one gets by not assuming it.

3 FLAT CURVES: CLOSED FORMS

Let $r_k = r$, $h_k = h$, and a regular schedule $\Delta_k = \Delta$ with $T = n\Delta$. Then $\mu = r + h$, $D_k = e^{-\mu k\Delta}$, and the sums are geometric:

$$\sum_{k=1}^{n} D_k = \frac{1 - e^{-\mu T}}{e^{\mu\Delta} - 1}, \qquad \sum_{k=1}^{n} \left(D_{k-1} - D_k\right) = 1 - e^{-\mu T}. \tag{TN25.10}$$

Substituting into (TN25.6) and (TN25.8),

$$A = \left(1 - e^{-\mu T}\right)\left[\frac{r\,\Delta}{\mu\,(e^{\mu\Delta} - 1)} + \frac{h}{\mu^2}\right], \qquad V = (1 - R)\,\frac{h}{\mu}\left(1 - e^{-\mu T}\right), \tag{TN25.11}$$

and the maturity factor $1 - e^{-\mu T}$ cancels in the ratio:

$$s = \frac{(1 - R)\,h}{\dfrac{h}{\mu} + \dfrac{r\,\Delta}{e^{\mu\Delta} - 1}}. \tag{TN25.12}$$

With flat curves the par spread does not depend on maturity, as a constant hazard rate requires.

Example 3.1 The chapters' numbers from the closed forms

Example 25.1 has $h = 0.025/0.6 = 4.1667\%$, $r = 3\%$, $R = 40\%$, quarterly premiums, five years, so $\mu = 7.1667\%$ and $\mu\Delta = 0.017917$. Equation (TN25.11) gives $A = 4.18652$ and $V = 10.5056\%$ of notional, and (TN25.12) gives $s = 250.94$ bp—the figures the example reports for protection paid at the default time. The contract of Example 4.1 has $h = 1\%$ with the same r and R; (TN25.12) gives 60.23 bp against the triangle's 60.00, the 0.376% that example builds one quarter at a time.

The protection buyer's position is concave in the spread. Section 25.4 states that a protection buyer's value, $M = (s - c)A$ for a contract paying the coupon c when the par spread is s, is concave in s. With flat curves this can be shown for every spread. Take the continuous-premium limit of (TN25.11), in which $A(\mu) = (1 - e^{-\mu T})/\mu$ and $s = (1 - R)h = (1 - R)(\mu - r)$. Then

$$M = (1 - R)\left(1 - e^{-\mu T}\right) - \left((1 - R)\,r + c\right)A(\mu), \tag{TN25.13}$$

because $\mu A(\mu) = 1 - e^{-\mu T}$. The first term is concave in μ, with second derivative $-(1 - R)T^2 e^{-\mu T}$. The annuity is convex, $A''(\mu) = \int_0^T t^2 e^{-\mu t}\,dt > 0$, and enters with a negative coefficient, so the second term is concave too. Since s is linear in μ,

$$\frac{d^2 M}{ds^2} = -\frac{1}{1 - R}\left[T^2 e^{-\mu T} + \left(r + \frac{c}{1 - R}\right)A''(\mu)\right] < 0$$

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at every spread, above the coupon or below it. The convexity of the annuity, which might seem to push the other way once $s > c$, is absorbed into $\mu A = 1 - e^{-\mu T}$, which is bounded by one: the premium stream's value saturates because default ends it sooner as the spread widens. With quarterly coupons the annuity is a finite sum and still convex, and μA is concave up to terms of order $(\mu\Delta)^2$, so the conclusion holds at any spread the chapter uses. Example 25.2's table shows it: over its five successive spread intervals the upfront rises by 0.0455, 0.0436, 0.0410, 0.0357 and 0.0287 percentage points per basis point. The seller's position, $-M$, is convex.

4 WHAT THE APPROXIMATIONS DO

The continuous-payment limit. As $\Delta \to 0$, $r\Delta/(e^{\mu\Delta} - 1) \to r/\mu$, so the denominator of (TN25.12) tends to one and

$$A \to \frac{1 - e^{-\mu T}}{\mu}, \qquad s \to (1 - R)\, h, \tag{TN25.14}$$

which is Section 24.2's credit triangle. For finite Δ the denominator $h/\mu + r\Delta/(e^{\mu\Delta} - 1)$ is slightly below one, so the discrete par spread sits slightly *above* $(1 - R)h$: expanding, the excess is $r\Delta/2$ to first order, which is the 0.376% of Example 4.1 at $r = 3\%$ and $\Delta = 0.25$. The reason is the one Figure 1 draws: a coupon paid at the end of a quarter is discounted for half a quarter longer, on average, than the same money paid continuously through it, while the protection is paid when default happens either way.

Example 4.1 The credit triangle is exact for a continuous premium, and the error people quote is the accrued premium

Price a five-year CDS with a flat hazard rate $h = 1\%$, discounting at $r = 3\%$ and recovery $R = 40\%$, three ways: with the premium paid continuously; quarterly, with the premium accrued since the last payment date paid on default, as real contracts do; and quarterly with no accrual. These are the inputs of Example 24.1 of Derivatives: Contracts, Models, and Markets, which prints only the resulting spreads. Here $\mu = r + h = 0.04$, so $D(t) = e^{-\mu t}$, and everything is per unit of notional.

The protection leg is the same in all three contracts:

$$(1 - R)\, h \int_0^5 e^{-\mu t}\, dt = 0.6 \times 0.01 \times \frac{1 - e^{-0.2}}{0.04} = 0.6 \times 0.01 \times 4.5317 = 0.027190.$$

The premium legs are the spread s times an annuity, and the annuities differ. One quarter is enough to see how, because with flat r and h every later quarter is the first one scaled by $D_{k-1} = e^{-\mu T_{k-1}}$. Over the first quarter $[0, \Delta]$ with $\Delta = 0.25$, a continuously paid premium collects $s\, dt$ at every instant the issuer survives,

$$s \int_0^{\Delta} e^{-\mu t}\, dt = s\, \frac{1 - e^{-0.01}}{0.04} = 0.248754\, s,$$

while a quarterly premium pays $s\Delta$ at the quarter end if the issuer has survived, with present value $s\Delta\, e^{-r\Delta} e^{-h\Delta} = s\Delta\, e^{-\mu\Delta}$, and, on default inside the quarter at time t, with probability $he^{-ht}\, dt$, the premium accrued so far, $s\, t$, whose present value is $s\, h\, t\, e^{-\mu t}\, dt$. Adding the survivors' payment to the default payments integrated over the quarter, which

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is (TN25.4) with $D_0 = 1$:

$$s\Big[\underbrace{\Delta\, e^{-\mu\Delta}}_{\text{quarter-end premium}} + \underbrace{h\int_0^{\Delta} t\, e^{-\mu t}\, dt}_{\text{accrual on default}}\Big] = (0.247512 + 0.000310)\, s = 0.247823\, s,$$

and without the accrual only the first term, $0.247512\, s$. The twenty scaling factors sum to $\sum_{k=0}^{19} e^{-0.01k} = 18.2177$, so over the whole contract the three premium legs are $4.5317\, s$, $4.5148\, s$ and $4.5091\, s$. The par spread sets each equal to the protection leg:

Continuous premium	$4.5317\, s = 0.027190$	$\Rightarrow\ s = 60.00$ bp
Quarterly, with accrual	$4.5148\, s = 0.027190$	$\Rightarrow\ s = 60.23$ bp
Quarterly, no accrual	$4.5091\, s = 0.027190$	$\Rightarrow\ s = 60.30$ bp

The first is $(1 - R)h = 0.6 \times 0.01$ exactly, as (TN25.14) says it must be. The second is higher by $4.53173/4.51477 = 1.00376$, or 0.376% of the spread, against the first-order $r\Delta/2 = 0.375\%$: Figure 1 shows the two payment patterns for a default inside the third quarter. The third drops the accrued premium, and that is the "approximation error" of the textbook statement: a tenth of a percent here, the annuity falling from 4.5148 to 4.5091. It is the value of a contract feature, not an error in the triangle, and unlike the 0.376% it grows with the hazard rate. Repeat the calculation at $h = 20\%$ and the three spreads are 1200.00, 1204.47 and 1235.17 bp, the quarterly contract still 0.37% above the triangle but the missing accrual now worth 2.5%, because twenty times as many issuers default inside a quarter.

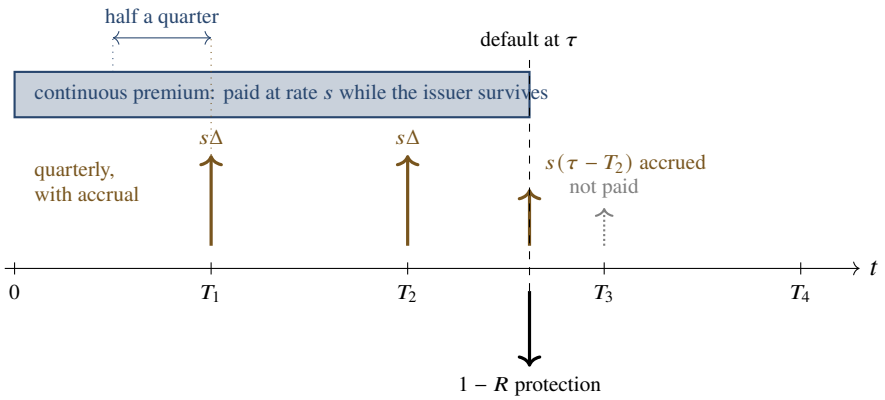


Figure 1: The premium leg of a CDS for a default at τ: paid continuously at rate s until default, or quarterly as $s\Delta$ at each quarter end plus the premium accrued to τ. The first quarter's continuous premium arrives, on average, half a quarter before the quarterly payment that replaces it. Schematic.

The zero-rate case. With $r = 0$, $\mu = h$ and (TN25.12) reduces to $s = (1 - R)\, h$ exactly, for every Δ: the accrued coupon paid at default exactly offsets the coupon lost on the default period. The tanh formula does not appear here. It arises only when the accrual term is replaced by the half-period rule. With $r = 0$ and accrual $\frac{1}{2}\Delta$ on default,

$$A_{\text{half}} = \left(1 - e^{-hT}\right)\left[\frac{\Delta}{e^{h\Delta} - 1} + \frac{\Delta}{2}\right] = \frac{\Delta\,(1 - e^{-hT})}{2} \cdot \frac{e^{h\Delta} + 1}{e^{h\Delta} - 1} = \frac{\Delta\,(1 - e^{-hT})}{2} \coth\!\left(\frac{h\Delta}{2}\right),$$

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and with $V = (1-R)(1-e^{-hT})$ the maturity factor cancels again:

$$s_{\text{half}} = \frac{2(1-R)}{\Delta}\tanh\Big(\frac{h\Delta}{2}\Big). \tag{TN25.15}$$

Since $\tanh x < x$, this sits slightly *below* the exact $(1-R)h$, with relative error about $h^2\Delta^2/12$, and recovers the triangle as $\Delta \to 0$. Note where the tanh belongs: in the numerator. The expression $2(1-R)/\big(\Delta\tanh(h\Delta/2)\big)$, which is sometimes written down by mistake, diverges like $4(1-R)/(h\Delta^2)$ as $\Delta \to 0$. For $r > 0$, with protection still paid at default, the half-period rule gives

$$s_{\text{half}} = \frac{(1-R)\,\dfrac{h}{\mu}\left(e^{\mu\Delta}-1\right)}{\dfrac{\Delta}{2}\left(e^{h\Delta}+1\right)},$$

which does not collapse to a single hyperbolic function. Restoring the tanh form for every r takes one more change of convention.

Default at mid-period, payment at the period end. Suppose that a default inside $(T_{k-1}, T_k]$ is deemed to occur at the midpoint, but that both the protection payment and the accrued coupon are settled at T_k. Writing $P_k = P(T_k)$ and $Q_k = Q(T_k)$, the unconditional probability of default in period k is $Q_{k-1} - Q_k$ and it is discounted with P_k, so

$$\begin{aligned} V &= (1-R)\sum_{k=1}^{n} P_k\big(Q_{k-1}-Q_k\big), \\ A &= \sum_{k=1}^{n}\Delta_k P_k Q_k + \sum_{k=1}^{n}\frac{1}{2}\Delta_k\, P_k\big(Q_{k-1}-Q_k\big) = \sum_{k=1}^{n}\Delta_k\, P_k\,\frac{Q_{k-1}+Q_k}{2}. \end{aligned} \tag{TN25.16}$$

The coupon in each period is weighted by the average of the survival probabilities at its two ends, which is the form used in most practitioner pricers and the form Section 25.3 writes. With flat curves $Q_{k-1} - Q_k = Q_k(e^{h\Delta}-1)$ and $Q_{k-1} + Q_k = Q_k(e^{h\Delta}+1)$, so every term is a multiple of $D_k = P_k Q_k$ and, using (TN25.10),

$$V = (1-R)\big(e^{h\Delta}-1\big)\frac{1-e^{-\mu T}}{e^{\mu\Delta}-1}, \qquad A = \frac{\Delta}{2}\big(e^{h\Delta}+1\big)\frac{1-e^{-\mu T}}{e^{\mu\Delta}-1}.$$

Both the maturity factor and the rate factor $e^{\mu\Delta}-1$ cancel in the ratio:

$$s = \frac{2(1-R)}{\Delta}\,\frac{e^{h\Delta}-1}{e^{h\Delta}+1} = \frac{2(1-R)}{\Delta}\tanh\Big(\frac{h\Delta}{2}\Big). \tag{TN25.17}$$

Under this convention the tanh formula is exact for every r and T, not only for $r = 0$, because default and payment are both tied to the period grid and the discount factor P_k multiplies the protection and coupon cash flows of period k identically. Expanding, $s = (1-R)\,h\left(1 - \frac{1}{12}h^2\Delta^2 + O(h^4\Delta^4)\right)$, slightly below the triangle: at Example 25.1's inputs it is 249.998 bp, the two thousandths of a basis point the chapter quotes. For $r = 0$ the annuity itself simplifies to $A = \frac{\Delta}{2}(1-e^{-hT})\coth(h\Delta/2)$.

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Why the half-period rule is nearly right. The accrued term of (TN25.16) can be set against the exact one (TN25.4) period by period. It carries two approximations, and they pull in opposite directions.

The first is the date. Conditional on default in period k, the elapsed time $u = \tau - T_{k-1}$ has density $h_k e^{-h_k u}/(1 - e^{-h_k \Delta_k})$ on $(0, \Delta_k]$, and integrating by parts gives

$$E\left[u \,\middle|\, T_{k-1} < \tau \le T_k\right] = \frac{1}{h_k} - \frac{\Delta_k e^{-x}}{1 - e^{-x}} = \Delta_k \left(\frac{1}{x} - \frac{1}{e^x - 1}\right), \qquad x = h_k \Delta_k. \quad \text{(TN25.18)}$$

The series $1/(e^x - 1) = 1/x - 1/2 + x/12 - x^3/720 + \cdots$, whose coefficients are the Bernoulli numbers, then gives

$$E\left[u \,\middle|\, T_{k-1} < \tau \le T_k\right] = \frac{\Delta_k}{2} - \frac{h_k \Delta_k^2}{12} + \frac{h_k^3 \Delta_k^4}{720} - \cdots$$

Default lands, on average, $h_k \Delta_k^2/12$ before the midpoint, a fraction $h_k \Delta_k/12$ of the period: the density falls across the period from h_k to $h_k e^{-x}$, so an early default is slightly more likely than a late one. Taken alone, this makes the rule's accrued coupon too large by a relative $h_k \Delta_k/6$.

The second is the discounting. The rule discounts the accrued coupon from T_k, but it is paid at τ, on average about half a period earlier. Expanding both terms to first order in Δ_k,

$$\int_{T_{k-1}}^{T_k} (t - T_{k-1})\, h_k\, D(t)\, dt = \frac{h_k D_{k-1} \Delta_k^2}{2} \left[1 - \frac{2}{3}(r_k + h_k)\Delta_k + O(\Delta_k^2)\right],$$

$$\frac{1}{2}\Delta_k\, P_k (Q_{k-1} - Q_k) = \frac{h_k D_{k-1} \Delta_k^2}{2} \left[1 - \left(r_k + \frac{h_k}{2}\right)\Delta_k + O(\Delta_k^2)\right],$$

so the exact accrual exceeds the rule's by a relative $(r_k/3 - h_k/6)\,\Delta_k$. The $-h_k/6$ is the earlier default date of (TN25.18); the $r_k/3$ is the discounting the rule overstates. At Example 25.1's inputs ($r = 3\%$, $h = 4.167\%$, $\Delta = 0.25$) the two are $+0.25\%$ and -0.17%, a net $+0.076\%$, against $0.001286635/0.001285647 - 1 = 0.077\%$ exactly (the first quarter's exact accrual against the rule's), which is the tenth of a percent Section 25.3 quotes. The two effects are of the same order and opposite sign, so they largely cancel, and to first order they cancel exactly when $h_k = 2r_k$.

The three conventions together. At Example 25.1's inputs the par spread is 250.94 bp when protection is paid at the default time and the accrual is integrated exactly (TN25.12), 250.94 bp under the half-period rule with protection still paid at default (250.939 against 250.938), and 249.998 bp when both payments are moved to the period end (TN25.17). The first two differ by the accrual approximation, which is worth less than a hundredth of a basis point at quarterly frequency; the last differs from them by the timing of the protection payment, which is worth about a basis point on a 250 bp contract and is the 0.376% of Example 4.1. All three tend to the credit triangle as $\Delta \to 0$, from above in the first two cases and from below in the third.

REFERENCES

O'Kane and Turnbull's "Valuation of Credit Default Swaps" (2003) is the practitioner derivation of the two legs and the source of the half-period rule as it is usually written;

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O'Kane's *Modelling Single-name and Multi-name Credit Derivatives* (2008) treats the accrual integral and the choice of payment timing at length. The ISDA CDS Standard Model, which the market uses to convert quoted spreads to upfronts, integrates the accrual within each period on a fine grid rather than assuming a midpoint, and Iwashita's OpenGamma note on single-name CDS conventions (2013) documents the schedule, day count, and coupon conventions the formulas above take as given.

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